DAY 04 · BEGINNER · FREE API

Chains: The Pipe

Three objects joined with one character, and data flowing left to right instead of inside out.

`|` `StrOutputParser` `.batch()` `.stream()` `Notebook`

KGP Talkie
LEARN. BUILD. TALK AI

Scan to subscribe

Laxmi Kant Tiwari · kgptalkie.com

The 10-Day Course Map

KGP Talkie
LEARN. BUILD. TALK AI

WEEK 1 TALKING TO A MODEL

- **01 Your First AI Program** — Ask a model, print the answer
- **02 Roles and Messages** — The system message sets the job
- **03 Prompt Templates** — One template, many inputs

YOU WILL HAVE BUILT
a script that answers anything you type into it

WEEK 2 BUILDING SOMETHING

- **04 Chains: The Pipe** — `prompt | model | parser` — YOU ARE HERE
- **05 Structured Output** — An object, not a paragraph
- **06 Memory** — A checkpointer and a thread id

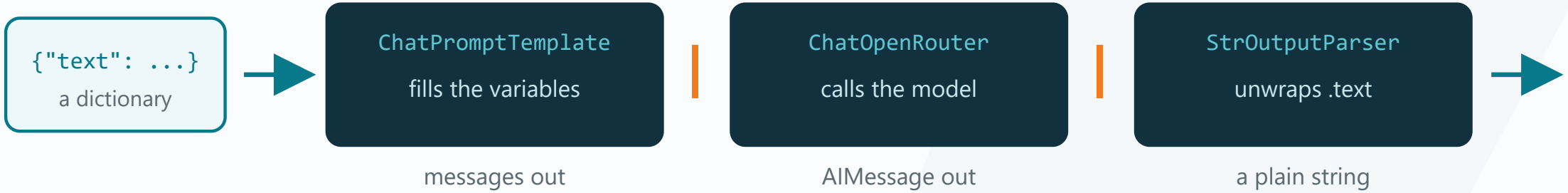
YOU WILL HAVE BUILT
a chatbot that remembers what you said a minute ago

WEEK 3 GIVING IT ABILITIES

- **07 Tools** — Your Python, called by the model
- **08 Your First Agent** — It decides when to call them
- **09 Chunks and Vectors** — Find a paragraph by meaning
- **10 RAG with RAGWire** — Citations, and a refusal

One notebook and one deck a day, **60 to 90 minutes** each. Every model used is a **free** one, and Docker only turns up on Day 9.

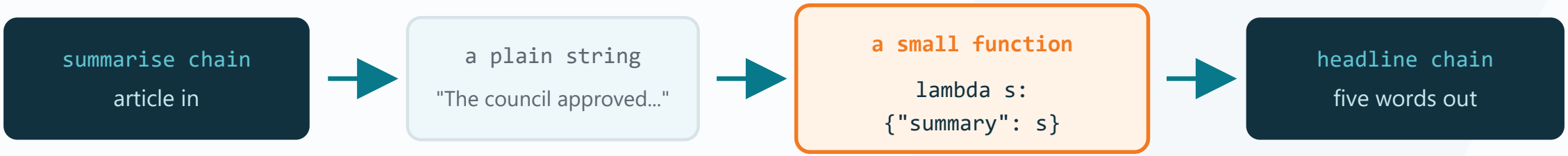
The Pipe: Template | Model | Parser



Not LangChain syntax: it is Python's or-operator, taught to pass output rightwards.

Read it **left to right**, not from the middle outwards.

Chaining Two Model Calls



a string came out of the left, a dictionary is needed on the right

When the shapes do not match, put a plain Python function in between.

A function you can read beats a clever expression you cannot.

.batch() Runs Many at Once

DEFAULT: CONCURRENT

article 1
article 2
article 3

→ all at once
fast

exactly what trips the per-minute limit
429 free-models-per-min

---

config={"max_concurrency": 1}

one, then the next, then the next → Slower, and it survives a room of twenty people

the whole class shares one free-tier limit

A big batch on a free key is the fastest way to make working code look broken.

Streaming the Reply

`.invoke()`

a blank screen, for four seconds

the whole answer

`.stream()`

The | council | approved | a plan | to plant | 12,000 trees...

The last word arrives at **the same moment** either way.

Nothing gets faster. The user just stops staring at nothing.

Tomorrow: Structured Output

WHAT YOU GET TODAY

```
chain.invoke({"text": review})

# 'This review is moderately
#  positive. The customer
#  praised the quality but
#  mentioned a delay...'
```

> Fine for a human. You cannot sort it, count it, or put it in a **database column**.

WHAT YOU GET TOMORROW

```
reader.invoke(review)

# ReviewScore(
#   rating=4,
#   positive=True,
#   reason='Excellent quality')
```

- An **object**, validated at the boundary.
- Fields you can constrain to a fixed set.
- And two quiet ways it fails, which you will meet.